Evaluating Protection Data Model Quality for Coordination Reviews

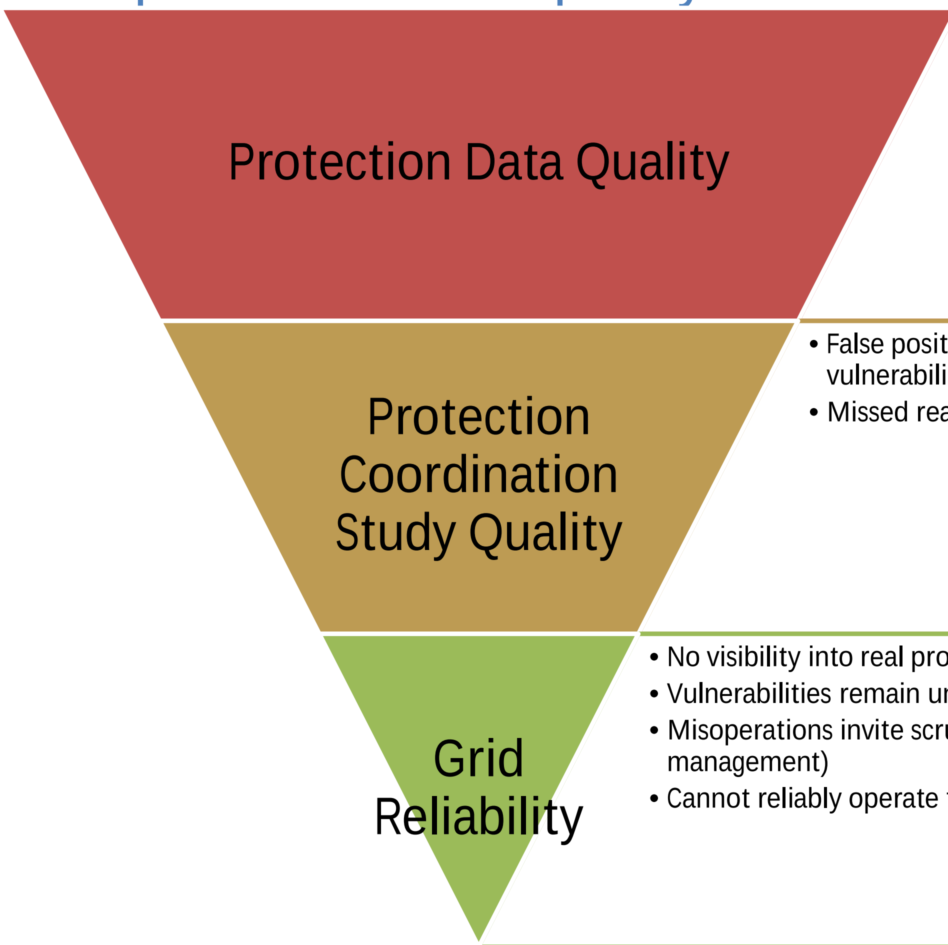
Gridscale X Advanced Protection Assessment
User Group Meeting
June 8-10, 2026

Agenda

- Protection data checking capabilities in APA
- Protection data checking with enhanced reporting
- Gridscale X APA macro and CSV output files
- Power BI reporting
- Planned improvements
- How can I get the macros and Power BI report?
- Q & A and Contact Information

Protection data checking capabilities in APA

Why perform a protection data quality check?

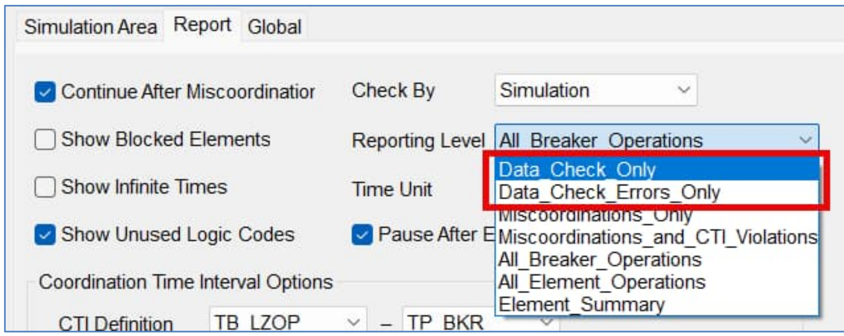


- Missing Operating and Polarizing CT/PT connections
- CT/PT connection mismatches
- Missing LZOP Logic
- Duplicate element logic codes
- Unused logic codes
- TOC math model errors
- Out of range settings

- False positives – not a real protection vulnerability
- Missed real protection vulnerabilities

- No visibility into real protection system issues
- Vulnerabilities remain unresolved
- Misoperations invite scrutiny from regulators (and management)
- Cannot reliably operate the grid

Protection data checking capabilities in APA




```
Simulation Area:
   130 buses;    141 branches
LINE LZOP                    90 lzops
MISC LZOP                     4 lzops
 94 distinct lzops
There are missing taps detected! Please check |ApaUserDir|\cape_15_log.txt for more details.
In Substation "SUB_8" LZOP 4396 LZOP_4396
LZOP #4396 has duplicated contact logic code "51G_1". Tripping suppressed for this LZOP

Errors/Warnings while Reading Data and Building Simulation
------------------------------------------------------------
There are missing taps detected! Please check |ApaUserDir|\cape_15_log.txt for more details.
In Substation "SUB_8" LZOP 4396 LZOP_4396
LZOP #4396 has duplicated contact logic code "51G_1". Tripping suppressed for this LZOP



                               TOC Curve Validation
                           -----------------------------

No TOC curve mismatches found.

Data verification completed
```

- Set Reporting Level to Data_Check_Only or Data_Check_Errors_Only
- Run a fault simulation in System Simulator or Relay Checking
- Summary output goes in the text area with details in log file

Presentation & review of the results of the data check – current process

Manual review

- Review text area in SS, RC or CM modules
- Review APA log file
- Limited to simulation area under consideration

Consolidate

- Consolidate data check runs from different studies
- Data quality issues for the same object may be reported in different studies
- Data issues for one asset reported in different areas of the same report

Data quality corrections

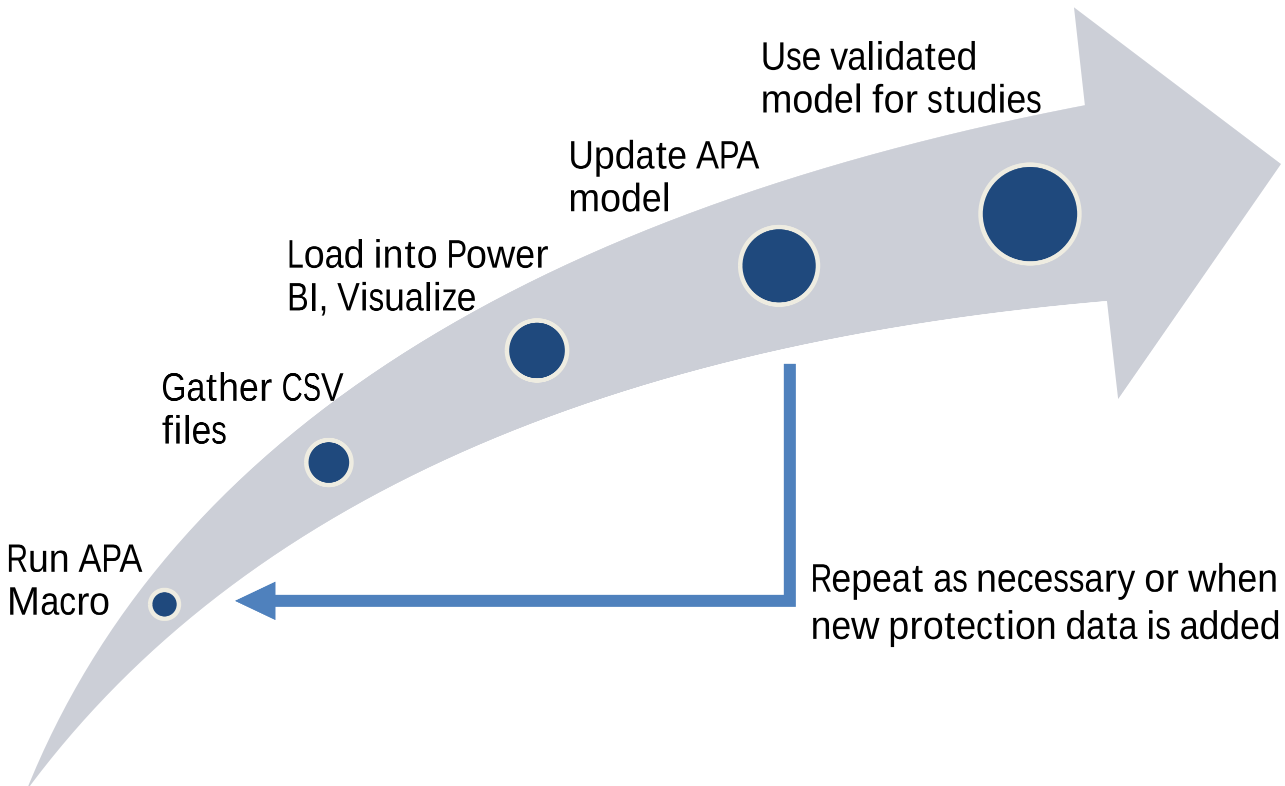
- Open Gridscale X APA and make necessary corrections/updates to the protection model

Requirement

- More automated data checking process
- Consolidation of disparate outputs
- Clear tabulation and visualization of data quality issues
- Unambiguous listing of actionable items

Protection data checking with enhanced reporting

New workflow summary



Detailed workflow

Gridscale X APA Macro

- Special data checking macro
- Run it on a single line or transformer, one substation or entire system
- Produces on-screen text output and CSV files

CSV output files

- Created by the new macro
- One CSV file for asset (line terminal or transformer) level information
- One CSV file for element-level information
- Each line of information in both CSV files is date-stamped

Power BI (PBI)

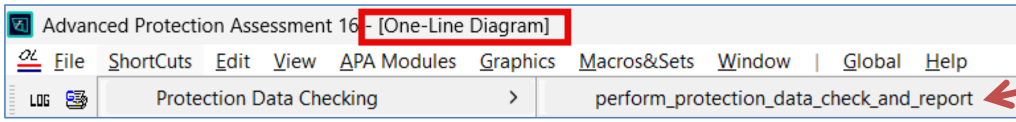
- Runs on the user's desktop
- Uses Microsoft's free licensing model for PBI desktop
- Manually ingests CSV files produced by the new macro
- Produces different types of visuals and charts to track protection data issues

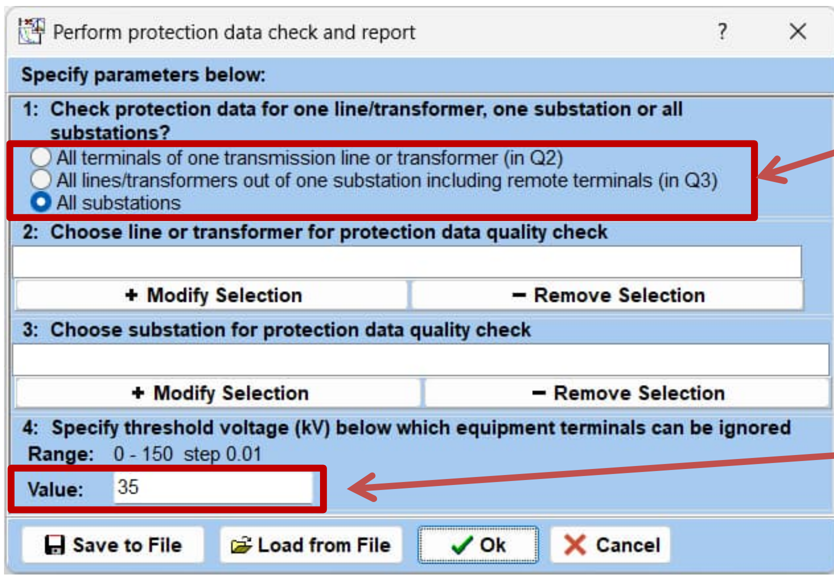
Gridscale X APA Model Updates

- Using PBI reports, update the APA model to fix data issues
- Repeat the process and verify via mitigation via PBI
- → Ready for protection coordination studies

Gridscale X APA macro and CSV output files

Gridscale X APA macro for checking protection data quality





- Run from APA OL Module ShortCuts menu
- Select:
  - One line or transformer (or)
  - One substation (or)
  - All substations in the model
- Specify cutoff voltage threshold below which equipment terminals will not be considered for the check

This macro is based on the existing "Auto Assign Contact Logic Codes from Functional Designations" Macro only for <u>retrieving element settings and displaying them on screen</u>.

Types of protection data checks performed by the macro

- Macro-based approach does not have access to internal data structures that the code-based checking tools have; the following data checks are performed

LZOP Level Checks

- Missing LZOP on a line or XFMR terminal
- Missing LZOP logic expression

Relay Level Checks

- Missing relays in LZOP
- Missing elements with contact logic codes

Element Level Checks

- Duplicate element logic codes within the LZOP
- Missing operating CT connection (IOC, TOC, DIST)
- CT location mismatch
- Missing operating VT connection (DIST)
- VT location mismatch

Text output on screen

Substation & LZOP information

Relay & element settings information

Element data issues

Substation: SUB_8 (Tag: 8)
XFMR: 57 BUS_57-12 BUS_12 ckt. 1 (Name: XFMR_7)
LZOP: LZOP_4924 TYPE: XFMR
LZOP Logic: DEFAULT_XFMR_TRIP

TAG	RELAY NAME	PROT. SCHEME	RELAY STYLE	CODE	DESIGNATION	ZONE	CTRATIO	PTRATIO	REACH PKP (A)	MTA	REACH%	TIMEDIAL OR DELAY	INVERSE TIME CHARAC.	TRIPCB	ELEMENT FUNCTION TYPE	CONTACT LOGIC CODE	DUPLICATELOGICCODE	CT MISSING?	CT BRANCH MISMATCH?	VT MISSING?	VT BRANCH MISMATCH?	VT SUB MISMATCH?
2614	RELAY_2656	PRIMARY	12BDD16B	CDIFF										Y	CDIFF	87DIFF_1	No					
2624	RELAY_2657	PRIMARY	12IAC53A2A	TOC			240.00		720			4.00	VERY INVERSE	Y	TOC_G	51G_1	No	No	Yes			

Substation: SUB_8 (Tag: 8)
XFMR: 57 BUS_57-12 BUS_12 ckt. 2 (Name: XFMR_6)
LZOP: LZOP_4923 TYPE: XFMR
LZOP Logic: DEFAULT_XFMR_TRIP

TAG	RELAY NAME	PROT. SCHEME	RELAY STYLE	CODE	DESIGNATION	ZONE	CTRATIO	PTRATIO	REACH PKP (A)	MTA	REACH%	TIMEDIAL OR DELAY	INVERSE TIME CHARAC.	TRIPCB	ELEMENT FUNCTION TYPE	CONTACT LOGIC CODE	DUPLICATELOGICCODE	CT MISSING?	CT BRANCH MISMATCH?	VT MISSING?	VT BRANCH MISMATCH?	VT SUB MISMATCH?
2616	RELAY_2658	PRIMARY	12BDD16B	CDIFF										Y	CDIFF							
2615	RELAY_2659	PRIMARY	12IAC53A2A	TOC			240.00		1440			3.00	VERY INVERSE	Y	TOC_G			No	No			

Substation: SUB_8 (Tag: 8)
LINE: 57 BUS_57-64 BUS_64 ckt. 1; Shortest path to remote bus: 65 BUS_65 @ 1.91313 ohms @ 79.7013 deg.
LZOP: LZOP_4396 TYPE: LINE
LZOP Logic: DEFAULT_LINE_TRIP

TAG	RELAY NAME	PROT. SCHEME	RELAY STYLE	CODE	DESIGNATION	ZONE	CTRATIO	PTRATIO	REACH PKP (A)	MTA	REACH%	TIMEDIAL OR DELAY	INVERSE TIME CHARAC.	TRIPCB	ELEMENT FUNCTION TYPE	CONTACT LOGIC CODE	DUPLICATELOGICCODE	CT MISSING?	CT BRANCH MISMATCH?	VT MISSING?	VT BRANCH MISMATCH?	VT SUB MISMATCH?
265	RELAY_236	PRIMARY	SEL-421-5_Z020	TOC	51S1T		300.00		750			2.30	U.S. Very Inverse	Y	TOC_G	51G_1	Yes	No	No			
265	RELAY_236	PRIMARY	SEL-421-5_Z020	DIST	M1P	1	300.00	1200.00	1.52000	80	79.449	0.00		Y	DIST_Z1_P	21P1_1	No	No	No	No	No	No
265	RELAY_236	PRIMARY	SEL-421-5_Z020	DIST	Z1G	1	300.00	1200.00	1.52000	80	79.449	0.00		Y	DIST_Z1_G	21G1_1	No	No	No	No	No	No
265	RELAY_236	PRIMARY	SEL-421-5_Z020	AUX	Z3D							42.00		Y	DIST_Z3_PG_T	21PG3T_1	No					
265	RELAY_236	PRIMARY	SEL-421-5_Z020	AUX	Z5D							24.00		Y	DIST_Z5_PG_T	21PG5T_1	No					
266	RELAY_237	SECONDARY	D60_v5.6x_5A	TOC	51N1		300.00		750			2.00	IAC VERY INV (D60)	Y	TOC_G	51G_1	Yes	No	No			
266	RELAY_237	SECONDARY	D60_v5.6x_5A	DIST	21G1	1	300.00	1200.00	1.52000	80	79.449	0.00		Y	DIST_Z1_G	21G1_2	No	No	No	No	No	No
266	RELAY_237	SECONDARY	D60_v5.6x_5A	DIST	21P1	1	300.00	1200.00	1.52000	80	79.449	0.00		Y	DIST_Z1_P	21P1_2	No	No	No	No	No	No
266	RELAY_237	SECONDARY	D60_v5.6x_5A	TIMER	T4_GND	1	300.00	1200.00	60	80	3136.1	42.00		Y	DIST_Z4_G_T	21G4T_1	No	No	No	No	No	No
266	RELAY_237	SECONDARY	D60_v5.6x_5A	TIMER	T4_PHS	1	300.00	1200.00	60	80	3136.1	42.00		Y	DIST_Z4_P_T	21P4T_1	No	No	No	No	No	No
266	RELAY_237	SECONDARY	D60_v5.6x_5A	TIMER	T5_GND	1	300.00	1200.00	2.60000	80	135.90	24.00		Y	DIST_Z5_G_T	21G5T_1	No	No	No	No	No	No
266	RELAY_237	SECONDARY	D60_v5.6x_5A	TIMER	T5_PHS	1	300.00	1200.00	2.60000	80	135.90	24.00		Y	DIST_Z5_P_T	21P5T_1	No	No	No	No	No	No

Text output on screen – element setting information

Relay & element settings information

Substation: SUB_8 (Tag: 8)
XFMR: 57 BUS_57-12 BUS_12 ckt. 1 (Name: XFMR_7)
LZOP: LZOP_4924 **TYPE: XFMR**
LZOP Logic: DEFAULT_XFMR_TRIP

TAG	RELAY NAME	PROT. SCHEME	RELAY STYLE	CODE	DESIGNATION	ZONE	CTRATIO	PTRATIO	REACH PKP (A)	MTA	REACH%	TIMEDIAL OR DELAY	INVERSE TIME CHARAC.
2614	RELAY_2656	PRIMARY	12BDD16B	CDIFF									
2624	RELAY_2657	PRIMARY	12IAC53A2A	TOC			240.00		720			4.00	VERY INVERSE

Substation: SUB_8 (Tag: 8)
XFMR: 57 BUS_57-12 BUS_12 ckt. 2 (Name: XFMR_6)
LZOP: LZOP_4923 **TYPE: XFMR**
LZOP Logic: DEFAULT_XFMR_TRIP

TAG	RELAY NAME	PROT. SCHEME	RELAY STYLE	CODE	DESIGNATION	ZONE	CTRATIO	PTRATIO	REACH PKP (A)	MTA	REACH%	TIMEDIAL OR DELAY	INVERSE TIME CHARAC.
2616	RELAY_2658	PRIMARY	12BDD16B	CDIFF									
2615	RELAY_2659	PRIMARY	12IAC53A2A	TOC			240.00		1440			3.00	VERY INVERSE

Substation: SUB_8 (Tag: 8)
LINE: 57 BUS_57-64 BUS_64 ckt. 1; Shortest path to remote bus: 65 BUS_65 @ 1.91313 ohms @ 79.7013 deg.
LZOP: LZOP_4396 **TYPE: LINE**
LZOP Logic: DEFAULT_LINE_TRIP

TAG	RELAY NAME	PROT. SCHEME	RELAY STYLE	CODE	DESIGNATION	ZONE	CTRATIO	PTRATIO	REACH PKP (A)	MTA	REACH%	TIMEDIAL OR DELAY	INVERSE TIME CHARAC.
265	RELAY_236	PRIMARY	SEL-421-5_Z020_	TOC	51S1T		300.00		750			2.30	U.S. Very Inverse
265	RELAY_236	PRIMARY	SEL-421-5_Z020_	DIST	M1P	1	300.00	1200.00	1.52000	80	79.449	0.00	
265	RELAY_236	PRIMARY	SEL-421-5_Z020_	DIST	Z1G	1	300.00	1200.00	1.52000	80	79.449	0.00	
265	RELAY_236	PRIMARY	SEL-421-5_Z020_	AUX	Z3D							42.00	
265	RELAY_236	PRIMARY	SEL-421-5_Z020_	AUX	Z5D							24.00	
266	RELAY_237	SECONDARY	D60_v5.6x_5A	TOC	51N1		300.00		750			2.00	IAC VERY INV (D60)
266	RELAY_237	SECONDARY	D60_v5.6x_5A	DIST	21G1	1	300.00	1200.00	1.52000	80	79.449	0.00	
266	RELAY_237	SECONDARY	D60_v5.6x_5A	DIST	21P1	1	300.00	1200.00	1.52000	80	79.449	0.00	
266	RELAY_237	SECONDARY	D60_v5.6x_5A	TIMER	T4_GND	1	300.00	1200.00	60	80	3136.1	42.00	
266	RELAY_237	SECONDARY	D60_v5.6x_5A	TIMER	T4_PHS	1	300.00	1200.00	60	80	3136.1	42.00	
266	RELAY_237	SECONDARY	D60_v5.6x_5A	TIMER	T5_GND	1	300.00	1200.00	2.60000	80	135.90	24.00	
266	RELAY_237	SECONDARY	D60_v5.6x_5A	TIMER	T5_PHS	1	300.00	1200.00	2.60000	80	135.90	24.00	

Text output on screen – data issues found

CT branch mismatch

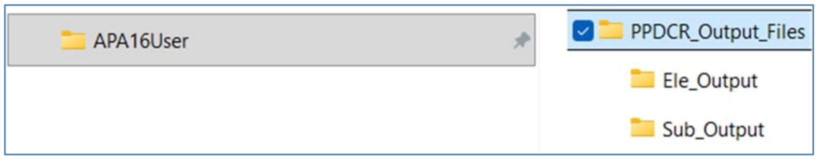
TRIPCB	ELEMENT FUNCTION TYPE	CONTACT LOGIC CODE	DUPLICATELOGICCODE	CT MISSING?	CT BRANCH MISMATCH?	VT MISSING?	VT BRANCH MISMATCH?	VT SUB MISMATCH?
Y	CDIFF	87DIFF_1	No					
Y	TOC_G	51G_1	No	No	Yes			

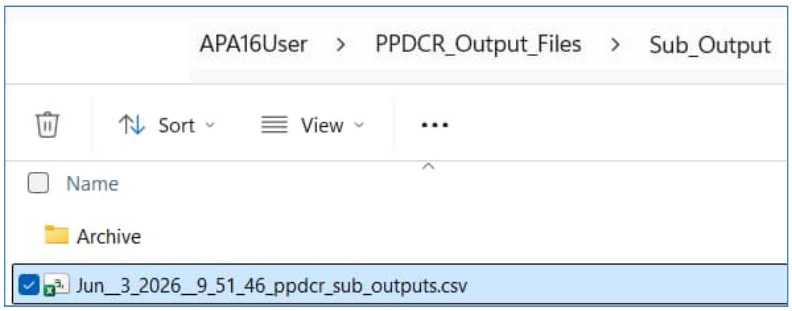
TRIPCB	ELEMENT FUNCTION TYPE	CONTACT LOGIC CODE	DUPLICATELOGICCODE	CT MISSING?	CT BRANCH MISMATCH?	VT MISSING?	VT BRANCH MISMATCH?	VT SUB MISMATCH?
Y	CDIFF							
Y	TOC_G			No	No			

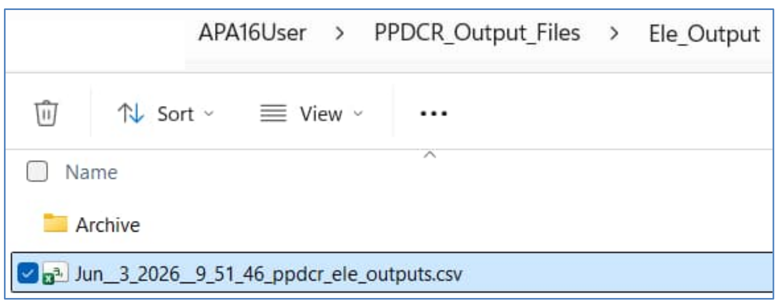
Duplicate contact logic codes

TRIPCB	ELEMENT FUNCTION TYPE	CONTACT LOGIC CODE	DUPLICATELOGICCODE	CT MISSING?	CT BRANCH MISMATCH?	VT MISSING?	VT BRANCH MISMATCH?	VT SUB MISMATCH?
Y	TOC_G	51G_1	Yes	No	No			
Y	DIST_Z1_P	21P1_1	No	No	No	No	No	No
Y	DIST_Z1_G	21G1_1	No	No	No	No	No	No
Y	DIST_Z3_PG_T	21PG3T_1	No					
Y	DIST_Z5_PG_T	21PG5T_1	No					
Y	TOC_G	51G_1	Yes	No	No			
Y	DIST_Z1_G	21G1_2	No	No	No	No	No	No
Y	DIST_Z1_P	21P1_2	No	No	No	No	No	No
Y	DIST_Z4_G_T	21G4T_1	No	No	No	No	No	No
Y	DIST_Z4_P_T	21P4T_1	No	No	No	No	No	No
Y	DIST_Z5_G_T	21G5T_1	No	No	No	No	No	No
Y	DIST_Z5_P_T	21P5T_1	No	No	No	No	No	No

CSV output files produced by the macro







- CSV files stored in APA16User folder under PPDCR_Output_Files
- Sub_Outputs file stores data at the level of the asset (line terminal or transformer) and provides a high-level summary of the issues found for that asset
- Ele_Outputs file stores detailed element-level data, as displayed on the screen

Power BI reporting

Power BI reporting

- Desktop-based Power BI report file
- Utilizes Microsoft's free desktop license

Line/XFMR terminal summary page

Top level summary

Line/XFMR terminal summary

Count of issues & distribution

Line/XFMR additional details

Line/XFMR terminal summary table

of Line/XFMR terminals with data issues

# of Line/XFMR terminals with high severity data issues

1564	464	454	29%	450	29%
Terminals	# of Issues	Terminals w/Issues	Terminals w/Issues (%)	Terminals w/High Sev. Issues	Terminals w/High Sev. Issues (%)

# of Line/XFMR terminals & total # of issues found

Last Update
04-Jun-2026, 15:06:36 (UTC)

Issue Severity

High: Missing LZOP, Logic, Relay, Elements with CLC, CT or VT issues
Low: Duplicate CLC or Elements with Trip CB = "Y" but no CLC
None: No protection data issues

Severity Selection

- ☑ High
- ☑ Low
- ☑ None

Select date range

6/3/2026 6/3/2026

Icon Legend

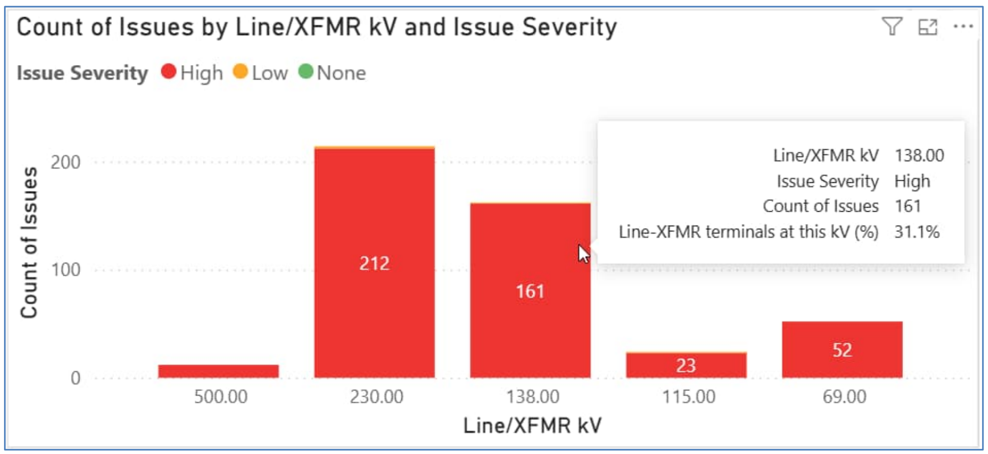
- ✓ Data Valid
- ✕ Data Issue
- ⚠ Data Missing
- — Not Applicable

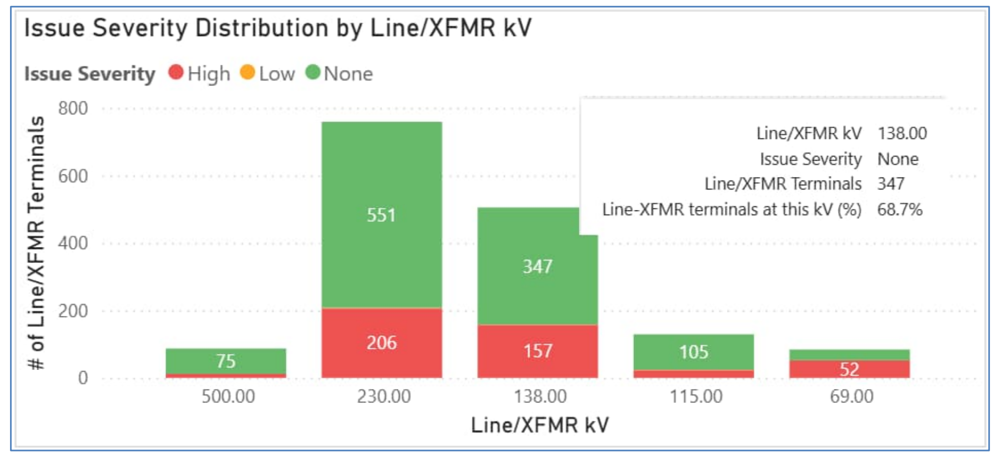
Issues per line/XFMR terminal

Line or Transformer Terminal Based Protection Data Quality Summary

Date	Line/XFMR Terminal ID	kV	Type	Substation	Issue Severity	LZOP Name	LZOP	Logic	Relays	Element CLC	Duplicate CLC	TripCB w/o CLC	CTs	VTs	Issue Count
03-Jun-2026	From 1140 BUS_1140 to 1147 BUS_1147 Circuit 1	500.00	LINE	SUB_813	High	LZOP_4222	✓	✓	⚠	—	—	—	—	—	1
03-Jun-2026	From 1140 BUS_1140 to 1148 BUS_1148 Circuit 1	500.00	LINE	SUB_813	High	LZOP_4224	✓	✓	✓	⚠	—	—	—	—	1
03-Jun-2026	From 1140 BUS_1140 to 1153 BUS_1153 Circuit 1	500.00	LINE	SUB_813	High	LZOP_4226	✓	✓	⚠	—	—	—	—	—	1
03-Jun-2026	From 1140 BUS_1140 to 1154 BUS_1154 Circuit 1	500.00	LINE	SUB_813	High	LZOP_4228	✓	✓	⚠	—	—	—	—	—	1
03-Jun-2026	From 1147 BUS_1147 to 1140 BUS_1140 Circuit 1	500.00	LINE	SUB_814	High	LZOP_4223	✓	✓	⚠	—	—	—	—	—	1
03-Jun-2026	From 1148 BUS_1148 to 1140 BUS_1140 Circuit 1	500.00	LINE	SUB_814	High	LZOP_4225	✓	✓	⚠	—	—	—	—	—	1
03-Jun-2026	From 1153 BUS_1153 to 1140 BUS_1140 Circuit 1	500.00	LINE	SUB_814	High	LZOP_4227	✓	✓	⚠	—	—	—	—	—	1
03-Jun-2026	From 1154 BUS_1154 to 1140 BUS_1140 Circuit 1	500.00	LINE	SUB_814	High	LZOP_4229	✓	✓	⚠	—	—	—	—	—	1
03-Jun-2026	From 1331 BUS_1331 to 1333 BUS_1333 Circuit 1	500.00	TCKT	SUB_862	High		⚠	—	—	—	—	—	—	—	1
03-Jun-2026	From 1331 BUS_1331 to 1334 BUS_1334 Circuit 1	500.00	TCKT	SUB_862	High		⚠	—	—	—	—	—	—	—	1
03-Jun-2026	From 1332 BUS_1332 to 1336 BUS_1336 Circuit 1	500.00	TCKT	SUB_862	High		⚠	—	—	—	—	—	—	—	1
03-Jun-2026	From 1332 BUS_1332 to 1337 BUS_1337 Circuit 1	500.00	TCKT	SUB_862	High		⚠	—	—	—	—	—	—	—	1

Line/XFMR terminals – count of issues and distribution

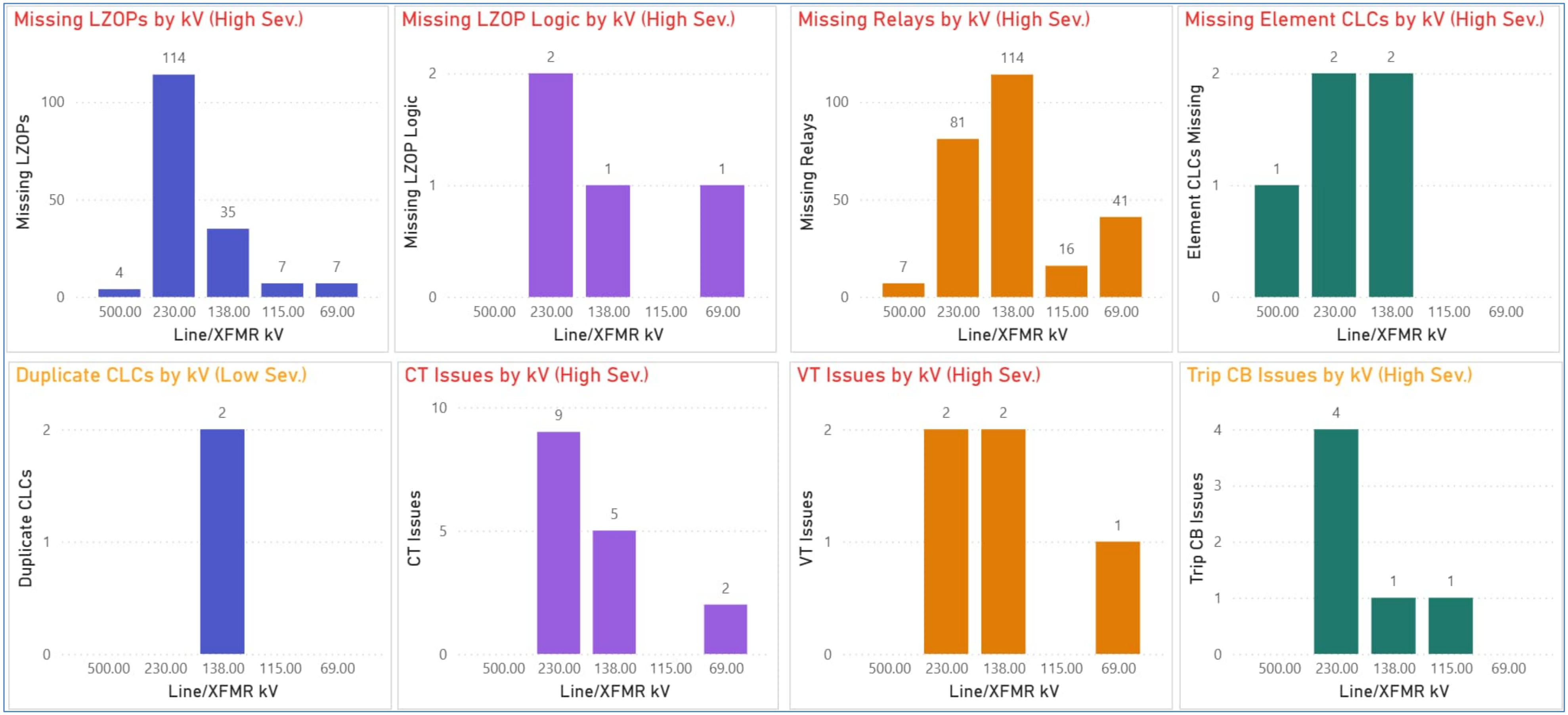




- Raw count of all issues found per voltage level
- High severity issues are predominant

- # of line/XFMR terminals per voltage level with high severity, low severity or no issues

Line/XFMR terminals – additional details



Drill through to element-level details

Line or Transformer Terminal Based Protection Data Quality Summary

Date	Line/XFMR Terminal ID	kV	Type	Substation	Issue Severity	LZOP Name	LZOP	Logic	Relays	Element CLC	Duplicate CLC	TripCB w/o CLC	CTs	VTs	Issue Count
03-Jun-2026	From 120 BUS_120 to 189 BUS_189 Circuit 1	500.00	LINE	SUB_94	None	LZOP_3639	✓	✓	✓	✓	✓	✓	✓	✓	0
03-Jun-2026	From 120 BUS_120 to 189 BUS_189 Circuit 2	500.00	LINE	SUB_94	None	LZOP_3641	✓	✓	✓	✓	✓	✓	✓	✓	0
03-Jun-2026	From 120 BUS_120 to 529 BUS_529 Circuit 1	500.00	LINE	SUB_94	None	LZOP_4498	✓	✓	✓	✓	✓	✓	✓	✓	0
03-Jun-2026	From 120 BUS_120 to 529 BUS_529 Circuit 2	500.00	LINE	SUB_94	None	LZOP_4500	✓	✓	✓	✓	✓	✓	✓	✓	0
03-Jun-2026	From 120 BUS_120 to 109 BUS_109 Circuit 1	500.00	TCKT	SUB_94	None	LZOP_4991	✓	✓	✓	✓	✓	✓	✓	—	0
03-Jun-2026	From 120 BUS_120 to 109 BUS_109 Circuit 2	500.00	TCKT	SUB_94	None	LZOP_4992	✓	✓	✓	✓	✓	✓	✓	—	0
03-Jun-2026	From 1830 BUS_1830 to 972		LINE	SUB_2770	High	LZOP_5055	✓	⚠	✓	⚠	—	✕	✕	✓	4
03-Jun-2026	From 1830 BUS_1830 to 463		LINE	SUB_2770	High	LZOP_5054	✓	⚠	✓	⚠	—	✕	✓	✓	3
03-Jun-2026	From 121 BUS_121 to 104		LINE	SUB_102	High	LZOP_4049	✓	✓	✓	✓	✓	✓	✕	✕	2
03-Jun-2026	From 1603 BUS_1603 to 1		TCKT	SUB_101	High		⚠	—	—	—	—	—	—	—	1
03-Jun-2026	From 1603 BUS_1603 to 2						⚠	—	—	—	—	—	—	—	1
03-Jun-2026	From 119 BUS_119 to 965					LZOP_4853	✓	✓	⚠	—	—	—	—	—	1

Copy
Show as a table
Include
Exclude
Drill through
Element-Level Details

- Right-click on any line of the Line/XFMR terminal summary table and drill through to the element-level details page

Element-level details for selected line or XFMR

Selected Substation
SUB_2770

Selected LZOP
LZOP_5055

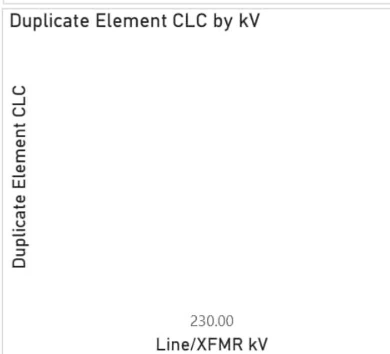
Select date range
6/3/2026 6/3/2026

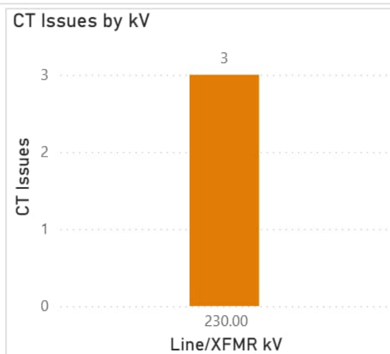
	LZOP Type	kV	Rly Tag	Rly Name	Scheme	Style	Code	Designation	Zone	CTR	PTR	TripCB	CLC	TripCB/CLC Mismatch	Duplicate CLC	CT Present	CT Locn. OK	VT Present	VT Locn. Ok	VT Sub. OK
5	LINE	230.00	2889	RELAY_2888	PRIMARY	SEL-421-5_Z022_5A	AUX	Z5D				Y		✕	—	—	—	—	—	—
5	LINE	230.00	2889	RELAY_2888	PRIMARY	SEL-421-5_Z022_5A	DIST	M1P	1	400.00	2000.00	Y		✕	—	✓	✕	✓	✓	✓
5	LINE	230.00	2889	RELAY_2888	PRIMARY	SEL-421-5_Z022_5A	DIST	Z1G	1	400.00	2000.00	Y		✕	—	✓	✕	✓	✓	✓
5	LINE	230.00	2889	RELAY_2888	PRIMARY	SEL-421-5_Z022_5A	TOC	51S1T		400.00		Y		✕	—	✓	✕	—	—	—

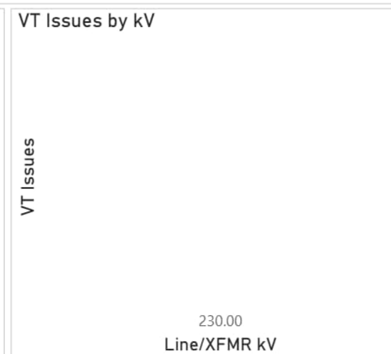
For selected substation and LZOP, element-level data issues are presented

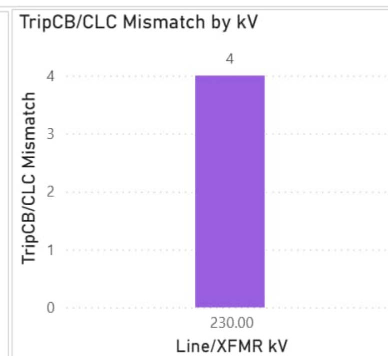
TripCB = Y, but no element CLC found

CT present, but possible location mismatch









Element-level details without any line/XFMR selection

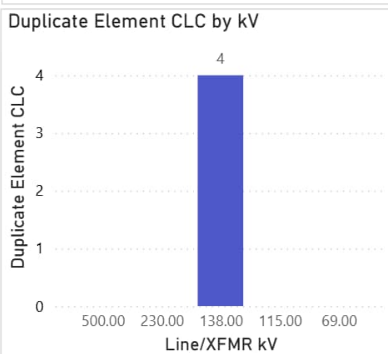
Selected Substation
Multiple / Not Selected

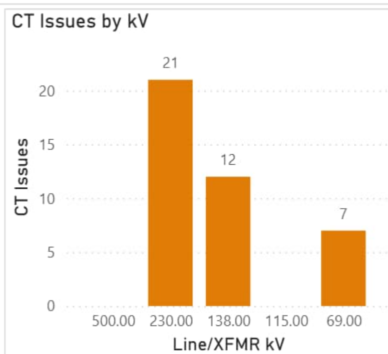
Selected LZOP
Multiple / Not Selected

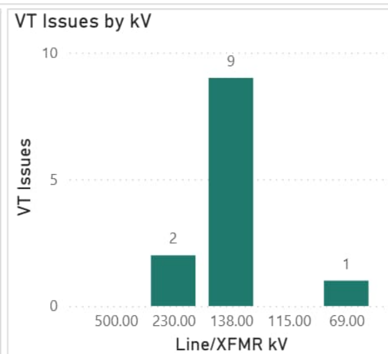
Select date range
6/3/2026 6/3/2026

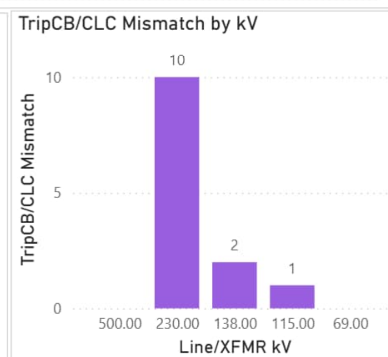
Several elements with no data issues

√	Rly Tag	Rly Name	Scheme	Style	Code	Designation	Zone	CTR	PTR	TripCB	CLC	TripCB/CLC Mismatch	Duplicate CLC	CT Present	CT Locn. OK	VT Present	VT Locn. Ok	VT Sub. OK
30.00	1910	RELAY_1903	SECONDARY	SEL-321-1_5A	DIST	M1P	1	400.00	2000.00	Y	21P1_1	✓	✓	✓	✓	✓	✓	✓
30.00	1910	RELAY_1903	SECONDARY	SEL-321-1_5A	DIST	Z1G	1	400.00	2000.00	Y	21G1_1	✓	✓	✓	✓	✓	✓	✓
30.00	1910	RELAY_1903	SECONDARY	SEL-321-1_5A	TIMER	Z2GD	1	400.00	2000.00	Y	21G2T_1	✓	✓	✓	✓	✓	✓	✓
30.00	1910	RELAY_1903	SECONDARY	SEL-321-1_5A	TIMER	Z2PD	1	400.00	2000.00	Y	21P2T_2	✓	✓	✓	✓	✓	✓	✓
30.00	1910	RELAY_1903	SECONDARY	SEL-321-1_5A	TOC	51N		400.00		Y	51G_1	✓	✓	✓	✓	—	—	—
30.00	211	RELAY_212	PRIMARY	12SAM11A22A	TIMER		1	400.00	2000.00	Y	21P2T_1	✓	✓	✓	✓	✓	✓	✓
30.00	1911	RELAY_1912	PRIMARY	LFZP111_5A	DIST	Z1	1	400.00	2000.00	Y	21PG1_1	✓	✓	✓	✓	✓	✓	✓
30.00	1911	RELAY_1912	PRIMARY	LFZP111_5A	TIMER	TZ2	1	400.00	2000.00	Y	21PG2T_1	✓	✓	✓	✓	✓	✓	✓
30.00	1911	RELAY_1912	PRIMARY	LFZP111_5A	TOC			400.00		Y	67GTOC_1	✓	✓	✓	✓	—	—	—
30.00	1912	RELAY_1913	SECONDARY	SEL-321-1_5A	DIST	M1P	1	400.00	2000.00	Y	21P1_1	✓	✓	✓	✓	✓	✓	✓
30.00	1912	RELAY_1913	SECONDARY	SEL-321-1_5A	DIST	Z1G	1	400.00	2000.00	Y	21G1_1	✓	✓	✓	✓	✓	✓	✓
30.00	1912	RELAY_1913	SECONDARY	SEL-321-1_5A	TIMER	Z2GD	1	400.00	2000.00	Y	21G2T_1	✓	✓	✓	✓	✓	✓	✓
30.00	1912	RELAY_1913	SECONDARY	SEL-321-1_5A	TIMER	Z2PD	1	400.00	2000.00	Y	21P2T_1	✓	✓	✓	✓	✓	✓	✓









Planned improvements

Planned improvements

Macro Improvements

- Restrict checking to a specific bus area
- Detect CT issues for CDIFF elements
- Detect Polarizing CT and VT issues for DIR elements
- Detect Logic code mismatches between element codes and tokens in LZOP logic expression
- CT summation point issues

Power BI Report Improvements

- Visualize new issues detected by the macro
- Provide ability to open the appropriate APA data form (LZOP, Relay, Element) from the report

How can I get the macros and Power BI report?

What do you need?

- There are three files that you will need:
  - Two macro (.mac) files
  - Power BI (.pbix) file

How can I get the files?

- Contact the presenter (details on last slide) to receive these files
  - Will provide instructions on installation and setup
  - No charge for the desktop version of these files

- Presenter will also make them available on GitHub (or another platform)
  - Bug fixes and updates are easily disseminated
  - Provides for better version control

Q & A and Contact Information

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